

Assignment 4

NIS3352-01 Modern Cryptography(2)

Marking Rubrics for Assignments.

- You will receive full marks if your idea is (mostly) correct and you haven't showed an adequate effort to present your ideas. No marks will be deducted if there are small errors or typos.
- If you are missing an important point of a problem, yet you have partially answered it, 10 marks of the question will be deducted.
- Full marks of the problem will be deducted if (1) your idea is not correct at all or (2) there is an obvious sign that you are copying it from your classmates.
- Each assignment is given one week to finish. Yet, we still accept it if it is at most one week late. For assignments two weeks after the deadline, 10 marks will be deducted.

Problem 1 (25 marks) Let $\{\mathbf{u}, \mathbf{v}\}$ be a basis for a lattice $\mathcal{L}$ in $\mathbb{R}^2$ where $\mathbf{u} = (3, 7)$ and $\mathbf{v} = (5, 10)$.

- Find the norm of $\mathbf{u}$ and the norm of $\mathbf{v}$.
- Determine the shortest vectors in $\mathcal{L}$.

Problem 2 (25 marks) Give a lattice $\mathcal{L} \subset \mathbb{R}^3$ with basis vectors:

$$\mathbf{v}_1 = (1, 1, 1), \mathbf{v}_2 = (1, 2, 3), \mathbf{v}_3 = (1, 0, 1),$$

perform the Gram-Schmidt orthogonalization on this set of basis vectors.

Problem 3 (25 marks) Prove that the "multi-secret" form of decision-LWE is hard, i.e., for any polynomial bounded t , tuples of the following form are indistinguishable from uniformly random:

$$(\mathbf{a}, b_1 = \mathbf{a}^T \cdot \mathbf{s}_1 + e_1, \dots, b_t = \mathbf{a}^T \cdot \mathbf{s}_t + e_t) \in \mathbb{Z}_q^n \times \mathbb{Z}_q^t,$$

where each $\mathbf{s}_j \leftarrow \mathbb{Z}_q^n$ is chosen once and for all, and the $\mathbf{a} \leftarrow \mathbb{Z}_q^n$ and $e_j \leftarrow \chi$ are independent. (Hint: use a hybrid argument: define a sequence of $t + 1$ distributions, where the first is the multi-secret LWE distribution and the last is uniformly random. Show that distinguishing any pair of adjacent distribution in the sequence is as hard as decision-LWE.)

Problem 4 (25 marks)

Dual Regev Public-Key Encryption Scheme. Let $n, m, q \in \mathbb{N}$, and let χ be an error distribution over $\mathbb{Z}_q$.

Key Generation.

1. Sample a matrix $A \xleftarrow{\$} \mathbb{Z}_q^{n \times m}$ uniformly at random.
2. Sample $s \in \{0, 1\}^m$ uniformly at random.
3. Compute

$$h := As \in \mathbb{Z}_q^n.$$

4. Output

$$\text{pk} = (A, h), \quad \text{sk} = s.$$

Encryption. To encrypt a bit $\mu \in \{0, 1\}$ under the public key $\text{pk} = (A, h)$:

1. Sample a vector $r \xleftarrow{\$} \mathbb{Z}_q^n$.

2. Sample error terms

$$e_1 \leftarrow \chi^m, \quad e_2 \leftarrow \chi.$$

3. Compute

$$\begin{aligned} c_1 &:= A^\top r + e_1 \in \mathbb{Z}_q^m, \\ c_2 &:= r^\top h + e_2 + \mu \cdot \left\lfloor \frac{q}{2} \right\rfloor \in \mathbb{Z}_q. \end{aligned}$$

4. Output the ciphertext

$$c = (c_1, c_2).$$

Decryption. Given a ciphertext $c = (c_1, c_2)$ and the secret key s , compute

$$d := c_2 - c_1^\top s \pmod{q}.$$

Since

$$c_2 - c_1^\top s = r^\top h + e_2 + \mu \cdot \left\lfloor \frac{q}{2} \right\rfloor - (A^\top r + e_1)^\top s,$$

we obtain

$$d = e_2 - e_1^\top s + \mu \cdot \left\lfloor \frac{q}{2} \right\rfloor \pmod{q}.$$

If the total error

$$\eta := e_2 - e_1^\top s$$

has magnitude much smaller than $q/4$, then:

- output $\mu = 0$ if d is closer to 0 than to $\lfloor q/2 \rfloor$ modulo q ;
- output $\mu = 1$ if d is closer to $\lfloor q/2 \rfloor$ than to 0 modulo q .

Prove that if the decision-LWE problem is hard, then the dual Regev encryption scheme is IND-CPA. (Hint: first, define game0, game1, and game2. Then, use the Leftover Hash Lemma and Hybrids argument to prove the indistinguishability between game0 and game1. After that, use LWE assumption to prove the indistinguishability between Game1 and Game2.)